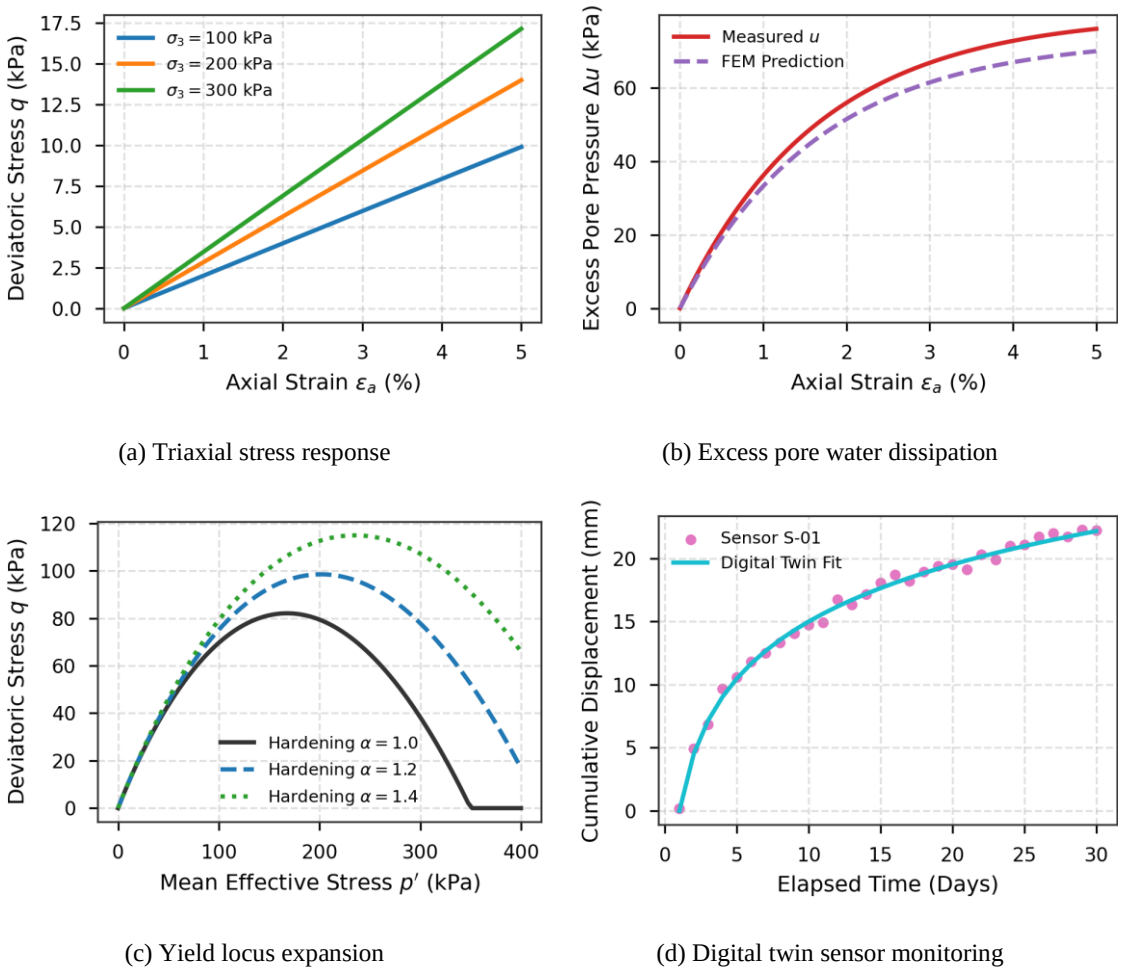
Here is an introductory paragraph describing the experimental setup.

Following the 2x2 panel, here is an evaluation of the two primary mechanisms.



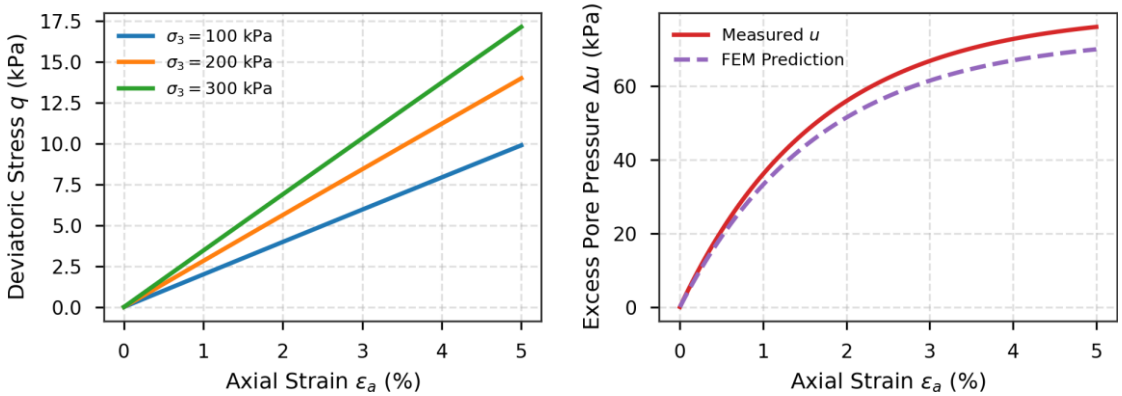
(a) Triaxial stress response

(b) Excess pore water dissipation

(c) Yield locus expansion

(d) Digital twin sensor monitoring

Fig. 1. Comprehensive mechanical and hydrological performance verification: (a) Stress-strain; (b) Pore pressure; (c) Yield locus; (d) Sensor tracking.



(a) Primary shear response (b) Dissipation kinetics

Fig. 2. Direct comparison between shear and dissipation behaviors.